when does the eclipse ide and its plugins migrate the an eclipse thea ide and extensions and what are the free open embedded and semiconductor ide's available

Researched Eclipse IDE migration and embedded IDEs

Eclipse IDE → Theia: No Formal Migration Exists

The classic **Eclipse IDE** (Java/SWT-based) and **Eclipse Theia** are **independent projects** under the Eclipse Foundation. There is no planned migration of the classic IDE to Theia, and the Foundation has explicitly stated:

> "This is not an announcement of the deprecation of the Eclipse IDE, the Eclipse Tool Platform or Eclipse RCP. These projects are stable, widely used, well maintained, and will continue to be so for a long time."

The classic Eclipse IDE is actively shipping — **v4.41** (Sept 2026), 63 participating projects, quarterly releases. It has no fixed EOL date.

What *is* happening is **individual RCP-based tools being rebuilt on Theia** by their respective vendors:

Tool	Vendor	Status
Code Composer Studio (CCS)	Texas Instruments	Rebuilt on Theia since v20 (D
STM32CubeMX2	STMicroelectronics	Built on Theia, launched Mar
Martini Designer	Lonti	Migrated from Eclipse RCP to
Sokatoa	Samsung	Built on Theia (2026)

The classic Eclipse IDE and its plugin ecosystem will **coexist** with Theia for the foreseeable future. The migration is bottom-up (individual tools choose to rebuild), not top-down (no Foundation mandate).

Free / Open-Source Embedded & Semiconductor IDEs

Built on Theia

IDE	Vendor	License	Notes
Code Composer Studio (CCS) v21	Texas Instruments	Free (proprietary)	Free for d simulators (Aug 2026
STM32CubeMX2	STMicroelectronics	Free (proprietary)	Graphical code gene 2026.
Mbed Studio	Arm	Apache 2.0	⚠️ **Mbed 2026** — S locally but maintaine

Eclipse-Based (classic platform)

IDE	Vendor	License	Notes
STM32CubeIDE	STMicroelectronics	Free (proprietary freeware)	v2.2.0 (Jun 20 dev: compile, Also has a VS

Fully Open-Source FPGA / Hardware

Tool	License	What it does
Aegis	Apache 2.0	**Full-stack open FPGA** (Apr 2026): fabric gene synthesis, P&R, bitstream, simulation, tapeout.
OpenCircuitX	MIT	HDL IDE (VHDL/Verilog) + visual circuit canva schematic viewer + waveform + FPGA toolcha ECP5). Windows now, Mac/Linux planned.

Icestudio	GPL-2.0	Visual block-based FPGA editor for open FPGA (iCE40, etc.).
Apio	GPL-3.0	CLI ecosystem for open FPGA boards — no toolchains/licenses/makefiles needed.
nextpnr	ISC	Vendor-neutral timing-driven FPGA place-and-
SymbiFlow	Various	Fully open-source FPGA toolchain (multiple ve
openFPGALoader	Various	Universal FPGA programming utility.
FOEDAG	BSD	Framework Open EDA GUI (Yosys-based).

In Progress

- **FEntwumS** (presented at FOSDEM 2026) — integrating the full open-source EDA toolchain (synthesis, P&R, bitstream) into **one coherent, vendor-agnostic, free IDE**. This is the closest thing to a "Vivado/Quartus replacement" in open source, but still early-stage.

Key Takeaway

The embedded/semiconductor space is **the primary growth domain for Theia** — TI, ST, and Arm all chose it. But the ecosystem is a **patchwork**: vendor tools (CCS, CubeMX2) are free-but-proprietary, while the truly open-source FPGA side (Aegis, OpenCircuitX, nextpnr) targets low-end/open hardware only. No single open-source IDE yet covers the full semiconductor workflow (ASIC tapeout remains gated by proprietary PDKs). The **FEntwumS** project is the most promising attempt to unify the open FPGA toolchain into a single IDE experience.

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what is the eta for FEntwumS candidate release and how does it compare with KiCad

Researched FEntwumS release and KiCad comparison

FEntwumS Release Status: No Published Candidate Date

There is **no announced "candidate release" date** for FEntwumS as a complete IDE. What exists today is active component-level development, not a cohesive product release.

What's actually shipping right now

The most active component is the **ContainerExtension** (Docker-based execution of open-source EDA tools inside OneWare Studio), which has been releasing rapidly in October 2026:

Version	Date	Notes
1.1.0	Oct 2, 2026	New interface from OneWare#331
1.1.1–1.1.6	Oct 5–9, 2026	Rapid iteration (bug fixes, workspace support)
1.2.0	**Oct 10, 2026**	Adds `docker.workspace`, removes a setting

This is a **Master's thesis project** (TH Köln, Summer 2026) that's feeding into the broader FEntwumS effort. The other components (SystemBuilder-GUI, SystemBuilder-Backend) have had **no commits since Feb 23, 2026**, and the OneWare fork itself was last updated Jun 9, 2026.

Realistic ETA estimate

Based on project signals:

- **Funding**: German BMBF research project (DI-FEntwumS via chipdesign-germany.de) — typical duration 3–4 years from start. If the project began ~2024, it likely runs through **2027–2028**.
- **Architecture**: Built on **OneWare Studio** (C#, GPL-3.0) as the host IDE — not Theia, not VS Code. This is a smaller ecosystem with fewer contributors.
- **Maturity**: The FOSDEM 2026 talk (Feb 2026) described the state as "working to integrate a whole range of these tools into one coherent IDE." Six months later, the integration is still at component-release stage, not a unified IDE build.

- **Best-case**: A **technical preview / alpha** of the full integrated IDE by **mid-2027**, assuming the thesis work stabilises and SystemBuilder development resumes. A **candidate release** (feature-complete, tested against the OpenEye case study) is more realistically **late 2027 or 2028**.

There is no public roadmap or release calendar. The project is research-driven, not product-driven.

FEntwumS vs KiCad: Different Domains, Complementary

They solve **orthogonal problems** in the hardware design flow. A direct "which is better" comparison doesn't apply — a complete FPGA board design uses **both**:

Dimension	KiCad (v9)	FEntwumS
Domain	Physical design: schematic, PCB layout, 3D, fab output	Logic design: HDL, synth simulation, bitstream
Maturity	v9.0 (Feb 2025), 20+ years of development, huge community	Research prototype, ~2 y single-institution team
Core tools	Schematic editor, PCB editor, ERC/DRC, Gerber/ODB++ export	Yosys (synthesis), nextpr GHDL (simulation), liteX
What it does NOT do	No HDL editing, no synthesis, no P&R, no bitstream	No PCB layout, no schen no fab output
FPGA relevance	Designs the **board** the FPGA sits on (power, connectors, clocking, constraints pinout)	Designs the **logic** inside (RTL → netlist → bitstrea
Licensing	GPL-2.0+	GPL-3.0 (OneWare) + va ISC, nextpnr ISC, GHDL
Platform	C++, cross-platform (Win/Mac/Linux)	C# (OneWare) + Docker cross-platform
User base	Hundreds of thousands of active users	Research consortium + e adopters

Where they intersect

The handoff point is the **pinout/constraint file**:

1. KiCad designs the PCB → exports a pinout/constraint file (e.g., XDC, SDC, or vendor-specific)
2. FEntwumS consumes that constraint → runs synthesis → P&R → bitstream
3. The bitstream is programmed onto the FPGA on the KiCad-designed board

FEntwumS's **NetlistViewer** (GHDL + Yosys + Avalonia) and **SystemBuilder** (liteX) are the components that would interface with KiCad's output. No formal KiCad plugin/integration exists yet.

The real comparison

If the question is "what's the open-source replacement for Vivado/Quartus?" — FEntwumS is the answer (eventually).

If the question is "what's the open-source replacement for Altium/OrCAD?" — KiCad is the answer (already shipping, mature).

For a **complete open-source FPGA board project**, you'd use:

```
KiCad 9 (PCB + schematic)
    ↓ pinout / constraints
FEntwumS (HDL → synthesis → P&R → bitstream)
    ↓
openFPGALoader / vendor JTAG (programming)
```

KiCad is production-ready today. FEntwumS is 1–2 years from production-ready, and even then it will be a **complement** to KiCad, not a replacement.

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so KiCad pcb design and FEntwumS FPGA logic gates and what is the free open tool for SoC SiP MCM PoP ASIC SoM CoM for circuits design

Researched open-source tools for SoC, ASIC, and packaging

The Full Stack: Open-Source Tools by Integration Level

The open-source landscape has a **critical cliff** at the packaging level. Everything below the die is well-served; everything at and above the package is almost entirely proprietary.

What exists (by domain)

Domain	Open-Source Tool	License
FPGA logic	FEntwumS (Yosys + nextpnr + GHDL + liteX)	ISC / EUPL / (
PCB / board	KiCad 9	GPL-2.0+
SoC / ASIC (RTL→GDSII)	OpenLane / LibreLane / SiliconCompiler	Apache-2.0
ASIC physical design	OpenROAD	BSD-3
RTL synthesis	Yosys	ISC
GDSII layout/view	KLayout	EUPL-1.2
Analog / mixed-signal ASIC	OpenFASoC (Xschem + ngspice + Magic)	GPL / EUPL
SoM / CoM	KiCad (same as PCB)	GPL-2.0+

The critical gap: SiP / MCM / PoP / 3D IC packaging

There is no open-source tool for this domain. The entire SiP/MCM/PoP design flow is locked behind proprietary licenses:

Function	Proprietary tools	Open-source equiv
Die-on-substrate placement	Cadence Allegro X APD, Synopsys IC Compiler II (3DIC)	**None**
BGA ball assignment / via stitching	Cadence, Zuken CR-8000	**None**
2.5D/3D interposer design	Synopsys, Cadence	**None**

Package-level SI/PI analysis	Ansys SIwave/Icepak, Siemens Calibre 3DSTACK	MEEP (partial — FD no package-specific DRC/routing)
Thermal (multi-die)	Ansys Icepak	Elmer FEM / OpenF (general-purpose, n specific)
Package substrate routing	Cadence, Zuken	**KiCad** (hackable — substrate as a very l count PCB, but no D package rules, no di constraints)

The closest you can get with open tools:

- **KiCad** for the substrate layout (it supports up to 60 layers, which covers most organic substrates)
- **MEEP** for electromagnetic simulation of the package structure
- **Elmer FEM / OpenFOAM** for thermal analysis
- **KLayout** for viewing the GDSII of individual dies before integration

But there is **no integrated flow** that takes multiple GDSII dies, places them on a substrate, assigns balls, stitches vias, and runs package-level SI/PI/thermal signoff. This is a **genuine gap** with no active open-source project I can find working to fill it.

SoM / CoM: no special tool needed

SoM and CoM design is **just PCB design in a constrained form factor**. The workflow is:

1. Choose a SoC (Raspberry Pi CM5, Toradex Verdin, Microchip SAMA5D27, etc.)
2. Design the carrier board in **KiCad** (power delivery, memory, connectivity, connectors)
3. Follow the vendor's pinout/mechanical spec

Open-source SoM/CoM projects that publish full KiCad source:

- **Raspberry Pi CM4 / CM5** — official KiCad files on GitHub
- **Microchip SAMA5D27 SOM** — open schematics, Gerber, BOM
- **OSM (Open Standard Module)** — open hardware + software standard (postage-stamp form factor, LGA, no connector)

- **Toradex** — reference carrier board schematics (not full KiCad, but detailed enough)

No dedicated "SoM design tool" exists or is needed — KiCad + the vendor's mechanical spec is the standard workflow.

Summary: where the open-source cliff is

FPGA logic	✅ FEntwumS (2027) / nextpnr / Yosys
SoC / ASIC	✅ OpenLane / SiliconCompiler / OpenROAD
PCB / SoM / CoM	✅ KiCad 9
SiP / MCM / PoP (die integration, substrate, 3D)	❌ NO OPEN-SOURCE TOOL (Cadence, Synopsys, Ansys, Zuken only)

The SiP/MCM/PoP layer is where the open-source ecosystem **completely breaks**. If you need to design a multi-die package (e.g., stacking a logic die with an HBM die on an interposer, or a PoP with DRAM over a base die), you are currently locked into **$50K–$200K/year proprietary tool licenses** with no viable open alternative. This is the single largest gap in the entire open hardware design stack.

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Identified open-source SiP/MCM/PoP gaps

The Gap Is Being Filled — Right Now, But Fragmentedly

The situation is **not** "nothing is happening." Multiple projects launched in 2025–2026 are directly targeting the SiP/MCM/PoP gap and the interoperability problem. But they're scattered across different foundations, funding bodies, and geographic regions, and **no one is building the unifying IDE or the full-stack handoff standard**.

What's actually in motion

Project	What it does	Status
IHP Open ADK	**First open-source Assembly Design Kit** for 2.5D heterogeneous chiplet integration. Places pre-existing dies, designs interposer/interconnect, 3D viewer, `.chiplet` format, manufacturable on IHP SG13G2 + IntM4TM2 interposer	Released **July 2026**
ODE4EC-DIG	Digital SoC design to GDSII, **explicitly includes "advanced packaging and die-to-die communication with open ADKs and physical demonstrators"**	Started **June 2026**, ru 2029
ODE4EC-PIV	Productivity, **interoperability** (modular frameworks, standardised interfaces, open data exchange formats), verification	Started **June 2026**, ru 2029
ODE4EC-AMS	Analog/mixed-signal open EDA (ngspice, ElmerFEM, Magic, KLayout-PEX)	Started **June 2026**, ru 2029
OCP Open Chiplet Economy	Vendor-neutral open framework for chiplet design, **Test Package PoC** (multi-vendor die integration), virtual prototyping, BoW (Book of Wormholes) die-to-die interface	Active workstreams, **C PoC** underway
ORFSW 2025	IEEE workshop identifying priority standards: intermediate representations, constraint/timing formats, exchange specs, **tool interoperability APIs**	Held **Nov 2025**, pape published
CHOPS	Open end-to-end design flow for **photonic + electronic** systems, explicitly covering	Launched **March 202**

	"development, **packaging**, and manufacturing" in open-source	
CDX (Chiplet Design Exchange)	Open-source, EDA-neutral, machine-readable **3DKs** (3D Design Kits) and chiplet authoring tools for 3D IC integration	Active within OCP

The **IHP Open ADK** is the single most important recent development. It's the first open-source tool that does what Cadence Allegro X APD and Synopsys IC Compiler II (3DIC) do for the 2.5D layer — and it's tied to a real, already-fabricated interposer.

The IHP team explicitly stated:

> "Heterogeneous integration is still taking shape: many of the approaches and standards for 2.5D and 3D chiplet systems are not yet settled. Here there is a genuine opening for the open-hardware community to study what already exists and **shape the standards while they are being defined**."

Foundation alignment: who's doing what

Foundation / Body	Role in this space	What they bring
FOSSi Foundation	**Anchor** for open EDA. Hosts ODE4EC, maintains Open PDKs / Ciel, runs ORConf	Community coordinat management, advoca roadmap
Linux Foundation	Hosts **OCP** (Open Compute Project), which runs the Open Chiplet Economy, CDX, and BoW standards	Industry consortium, body, multi-vendor pa (Intel, AMD, ARM, etc
Eclipse Foundation	Hosts **Theia** (the IDE framework), Eclipse AI Registry	The **IDE layer** — the that could unify the fu
IEEE	ORFSW workshop, CASS Standards Activities	Standards process, a industry bridge
EU Chips JU	**Funding body** for ODE4EC (3 projects, ~2026–2029)	The actual money
IHP (Leibniz)	Built the Open ADK, open PDK (SG13G2), MPW shuttle	The only European fo an open PDK + open

F-Si (Free Silicon Foundation)	Advocacy, copyleft licensing for EDA, Guix packaging, policy recommendations	Philosophical directic
Apache Foundation	**No specific hardware/EDA project**	—
FSF / GNU	**No specific hardware/EDA project** (F-Si is independent)	—

What's still missing — the actual gaps

1. No unified IDE for the full stack

Every tool in the chain is a **CLI or script-based flow**. There is no Theia-based (or any other) IDE that lets a designer go from RTL → synthesis → P&R → GDSII → package assembly → PCB layout in one environment. The Eclipse Foundation hosts Theia, and Theia has VS Code extension compatibility, but **no one has built the full-stack EDA IDE**. This is the single largest "missing project" — and it's a natural fit for Theia's architecture (customizable workbench, web + desktop, dependency injection for replacing any component).

2. No standardized handoff format between layers

The current handoffs are:

- RTL → GDSII: LEF/DEF + GDSII (well-standardized)
- [illegible] CDX **3DK** (OCP, still maturing)
- Package → PCB: **No standard**. KiCad doesn't natively consume GDSII or `.chiplet` files. You manually translate pinouts.
- PCB → Fab: Gerber/ODB++ (well-standardized)

The **package-to-PCB handoff** is the weakest link. ODE4EC-PIV's interoperability work is targeting this, but it's a 3-year project that just started.

3. No open-source package-level SI/PI/thermal signoff

IHP's ADK does placement and interconnect design. But the **electromagnetic simulation, signal integrity, power integrity, and thermal analysis** of the assembled package still requires Ansys SIwave/Icepak or Siemens Calibre. MEEP and Elmer FEM exist but are general-purpose FDTD/FEM solvers — they don't have

package-specific DRC, extraction, or signoff workflows. This is a genuine gap with no active project.

4. Funding is heavily EU/German-centric

Region	Funding mechanism	Coverage
EU	Chips JU → ODE4EC (3 projects, 2026–2029)	AMS, Digital, PIV — **no dedica packaging project**
Germany	BMBF → CHOPS, FEntwumS, DI-FEntwumS	Photonic integration, FPGA
US	DARPA → OpenROAD (ended), CHIPS Act (mostly fabs, not EDA)	RTL-to-GDSII only
UK	**No structured open-source EDA funding program**	—
Asia	Japan (Rapidus, Rapidus open PDK discussions), South Korea (not public)	Early stage

The UK has **no equivalent** to the EU Chips JU or BMBF programs for open-source EDA. This is a notable gap given the user's location.

5. No foundation is coordinating the full-stack picture

- FOSSi Foundation covers EDA tools (RTL-to-GDSII, AMS)
- OCP / Linux Foundation covers chiplet standards and die-to-die interfaces
- Eclipse Foundation hosts Theia but has no EDA-specific initiative
- No one is sitting at the table saying "here's the full flow from FPGA logic through ASIC to package to PCB, here are the handoff points, here's the IDE, here's the funding plan"

The **acatech IMPULSE report** (Jan 2026, German National Academy of Sciences) explicitly calls for "open-source assembly design kits (ADKs)" and identifies the APECS pilot line as the fabrication backbone, but it's a policy report, not a project.

The missing project: what would need to be built

If you were to propose the project that fills the remaining gaps, it would look like this:

Component	What it is	Natural home
Full-stack EDA IDE	Theia-based workbench with panels for RTL, synthesis, P&R, GDSII viewing (KLayout), package assembly (IHP ADK), PCB (KiCad), and simulation	Eclipse Foundation FOSSi Foundation
Package-to-PCB bridge	Open-source converter from `.chiplet` / 3DK / GDSII → KiCad project (pinout, mechanical outline, thermal pads, power domains)	KiCad project + IHP
Open package SI/PI/thermal signoff	Package-specific EM extraction + SI/PI analysis + thermal, built on MEEP/ElmerFEM with a package-aware DRC engine	New project, BMBF funding
Unified handoff standard	A single open data format (extending `.chiplet` / 3DK) that carries die + package + board-level information through the full flow	OCP / IEEE CASS (follow-up)
UK/US funding mechanism	A structured program (analogous to ODE4EC) for open-source EDA in the UK and US	UKRI / US CHIPS A center funding

Verdict

The SiP/MCM/PoP gap is **no longer unaddressed** — IHP's Open ADK (July 2026) and ODE4EC (June 2026–2029) are the first concrete, funded, open-source responses. The interoperability work is being driven by ODE4EC-PIV and ORFSW. But the **IDE layer**, the **package-to-PCB handoff**, and **package-level signoff** are still empty. And the coordination is fragmented across four foundations (FOSSi, Linux/OCP, Eclipse, IEEE) with no single entity owning the full-stack picture. The acatech report and the EU Chips Act provide the policy direction, but the engineering work is 2–3 years out at best.